

Curriculum Vitae

Xiangqian Yang

Contact Information

Beijing International Center for Mathematical Research,
Peking University,
No. 5 Yiheyuan Road, Haidian District,
Beijing, 100871, China.
Email: yangxq@pku.edu.cn

Education

- Ph.D. student, in Mathematics Science, **Peking University**, September 2022 – Present. Advisor: Liang Xiao.
- B.S. in Mathematics Science, **Peking University**, September 2018 – July 2022.

Other Experiences

- Visiting Student Researcher, **Stanford University**, November 2024 – January 2025, Mentor: Xinwen Zhu.
- Visiting Student Research Collaborator, **Princeton University**, October 2023 – June 2024. Mentor: Shou-Wu Zhang.

Research Interests

- Langlands Program.
- Arithmetic Geometry

(Pre)publications

1. On the generic part of the cohomology of Shimura varieties of abelian type (joint with Xinwen Zhu). *arXiv:2505.04329*.

2. On Ihara's lemma for definite unitary groups. *arXiv:2504.07504*.

Research Talks

- Sino-French Number Theory Workshop, Tianyuan Mathematics Research Center, Kunming, July 2026.
- Categorical Langlands Summit, KIAS, Seoul, June 2026.
- Workshop on Algebra and Number Theory 2026, Sichuan University, online talk, June 2026.
- Seminar on Langlands and Arithmetic Geometry, National University of Singapore, April 2026.
- Number Theory Seminar, The University of Chicago, online talk, February 2026.
- Number Theory Seminar, Cambridge University, October 2025.
- Number Theory Seminar, Oxford University, October 2025.
- BIRS-CMO workshop on “Categorical and Geometric Structures in the Langlands Program”, Oaxaca, online talk, September 2025.
- BIRS-IASM workshop on “New Tools in the Local and Global Langlands Program”, Hangzhou, July 2025.
- Minicourse in “Lectures on Arithmetic Geometry”, Peking University, June 2025.
- Number Theory and Representation Theory Seminar, Zhejiang University, June 2025.
- Number Theory Seminar, Stanford University, January 2025.
- Conference on “ p -adic Methods in Number Theory and Arithmetic Geometry”, Tianyuan Mathematics Research Center, Kunming, November 2024.

Teaching Experiences

- Teaching Assistant for “Advanced Mathematics B(II)”, Peking University, Spring 2026.
- Teaching Assistant for “Linear Algebra A(I)”, Peking University, Fall 2025.

- Teaching Assistant for “Number Theory II”, Peking University, Spring 2025.
- Teaching Assistant for “Abstract Algebra”, Peking University, Fall 2022.

Honors & Awards

- Huaixin Bachelor, Beijing International Center for Mathematical Research, Peking University, 2022.
- Two Gold Medals in the Chinese Mathematical Olympiad, 2016 and 2017.